

FLEX-DOT: END-TO-END DIFFERENTIABLE OPERATOR NETWORKS FOR SELF-EXTRACTING CONCEPT BOTTLENECK EXPERTS

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ABSTRACT

Concept Bottleneck Models (CBMs) predict a human-interpretable task from predicted concepts, and recent *Mixture of Concept Bottleneck Experts* (M-CBE) generalize the task predictor to a mixture of user-aligned symbolic expressions. Their most flexible instantiation, Symbolic M-CBE, recovers ground-truth equations but relies on a brittle three-stage pipeline—train a black-box predictor, distill it offline with genetic-programming symbolic regression, then substitute and fine-tune—that is slow, non-faithful to the trained network, and numerically unstable at test time (a distilled c_0/c_1 produces 8644 test MSE from near-zero denominators). We introduce FLEX-DOT, a single-stage, end-to-end differentiable operator network in which every expert is a differentiable mixture over six *complete canonical topology templates* (affine, product, bounded quotient, transcendental, harmonic, quadratic). An annealed straight-through selector commits each expert to exactly one template, so the argmax topology *is* the prediction and the closed-form equation is read verbatim out of the trained network with 100% faithfulness—no offline distillation. A bounded reciprocal $\text{safe_div}(a, b) = ab/(b^2 + \epsilon)$ removes the division blow-up ($439\times$ lower MSE than Symbolic M-CBE), and complete-in-family templates for periodic and distributive targets close the accuracy gaps of prior work. Across four regression benchmarks FLEX-DOT attains the best accuracy among all concept-transparent models—even surpassing an oracle given the ground-truth equations on three of four tasks—while recovering the exact data-generating mechanism (tree-edit distance 0) on three of four datasets with expressions $2\text{--}5\times$ more compact than a black-box mixture. Extensive ablations isolate each design choice.

1 INTRODUCTION

High-stakes applications increasingly demand models that are not merely accurate but *interpretable by construction* (Rudin, 2019; Rudin et al., 2022). Concept Bottleneck Models (CBMs) (Koh et al., 2020) answer this by routing every prediction through a layer of human-understandable concepts c predicted from the raw input x , and then predicting the task y from c . The transparency of a CBM, however, is only as good as the transparency of its *task predictor*: a linear map is globally inspectable but often too rigid, while a black-box MLP restores accuracy at the cost of interpretability.

Recent work on *Mixture of Concept Bottleneck Experts* (M-CBE) reframes the task predictor as a random variable over *expression trees*, conditioned on the input through a discrete selector, and shows that navigating two axes—the *functional form* of each expert and the *number of experts* M —traces out the accuracy–interpretability Pareto frontier (Barbiero et al., 2023; Debot et al., 2024). The most expressive instantiation, *Symbolic M-CBE* (Sym-M-CBE), discovers the concept-to-task equation itself, and in several cases recovers the exact data-generating mechanism (e.g. Pendulum’s $8 \sin \theta + 10$).

Sym-M-CBE achieves this through a **three-stage pipeline**: (i) train a mixture of black-box MLP experts jointly with the encoder and selector; (ii) run an *offline* genetic-programming symbolic-regression search (PySR (Cranmer, 2023)) on each expert’s inputs/outputs to recover a symbolic tree; (iii) substitute the discovered trees back and fine-tune. This decoupling is powerful but carries

three concrete costs. (1) *It is not faithful*: the extracted equation is fit to the MLP’s behaviour, not equal to what the network computes. (2) *It is unstable*: PySR readily distills a raw c_0/c_1 operator, which extrapolates to enormous errors whenever a predicted denominator drifts near zero—on MNIST-Arithmetic this inflates test MSE to 8644 (vs. 20 for a linear expert). (3) *It is slow and hard to tune*: an evolutionary search runs per expert, per dataset, outside the training loop.

Contributions. We propose **FLEX-DOT** (Flexible Topological Differentiable Operator Networks), which makes every M-CBE expert an *end-to-end differentiable operator network* and eliminates the offline distillation stage entirely. Our contributions are:

- **A single-stage self-extracting expert.** Each expert is a differentiable mixture over six *complete canonical topology templates* (Sec. 3). An annealed straight-through Gumbel selector (Jang et al., 2017; Bengio et al., 2013) commits it to exactly one template, so the *argmax topology is the prediction* and its SymPy read-out is exact (100% faithful)—no PySR (Fig. 1).
- **Numerical stability by design.** A bounded reciprocal $\text{safe.div}(a, b) = ab/(b^2 + \epsilon)$ removes the division blow-up that plagues distilled quotients, cutting MNIST-Arithmetic test MSE from 8644 (Sym-M-CBE) to 20—a $439\times$ improvement (Fig. 3b).
- **Complete-in-family templates** for periodic (harmonic) and multi-concept distributive (quadratic) targets close the two accuracy gaps of prior work, so FLEX-DOT is the most accurate concept-transparent model on all four regression benchmarks—beating even an oracle given the ground-truth equations on three (Fig. 2).
- **A thorough empirical study** of accuracy, exact structural recovery (Fig. 3a), expert cardinality (Fig. 4b), and five ablations isolating the safe-division bound, straight-through selection, structural priors, and coactivation.

2 BACKGROUND AND RELATED WORK

Concept-based models. CBMs (Koh et al., 2020) predict concepts c then a task $y = g(c)$; Concept Embedding Models (Espinosa Zarlenga et al., 2022) improve accuracy with concept embeddings at the price of an opaque predictor. Locally interpretable variants such as DCR (Barbiero et al., 2023) and LICEM (De Santis et al., 2025) generate a *per-sample* rule, which precludes global inspection because every input yields a new expression. CMR (Debot et al., 2024) restores global interpretability with a finite memory of Boolean rules, but conjunctive rules become brittle as the number of concepts grows. M-CBE unifies these under a probabilistic model in which the task predictor is a random expression tree T with $p(x, c, t, y) = p(x)p(c | x)p(t | x)p(y | c, t)$, and $p(t | x) = \sum_{m=1}^M p(t | m)p(m | x)$ is a mixture over a finite set of M experts selected by a learnable router. FLEX-DOT inherits this graphical model and this notion of global interpretability (a finite, inspectable set of experts), and contributes a new way to *realize each expert*.

Symbolic regression and differentiable operators. Distillation-based symbolic regression (Schmidt & Lipson, 2009; Cranmer et al., 2020; Cranmer, 2023) searches expression space with genetic programming, offline. Equation Learner networks (Martius & Lampert, 2017) and KANs (Liu et al., 2025) instead make the operator structure differentiable, but produce large, over-parameterized expressions requiring aggressive post-hoc sparsification. FLEX-DOT takes the differentiable route but replaces a free operator graph with a *small bank of complete canonical forms*, each a self-contained closed form, so a single hard selection yields a clean, exact equation. Our discrete selection uses straight-through Gumbel-Softmax (Jang et al., 2017; Maddison et al., 2017; Bengio et al., 2013); the finite mixture of experts follows the sparsely-gated routing of Shazeer et al. (2017), here specialized to interpretable canonical templates.

3 METHOD: FLEX-DOT

FLEX-DOT keeps the M-CBE generative model of Sec. 2 and replaces each black-box expert with a differentiable operator network whose closed form can be read out exactly. We first describe a single expert, then the selection mechanism, the structural priors, and training.

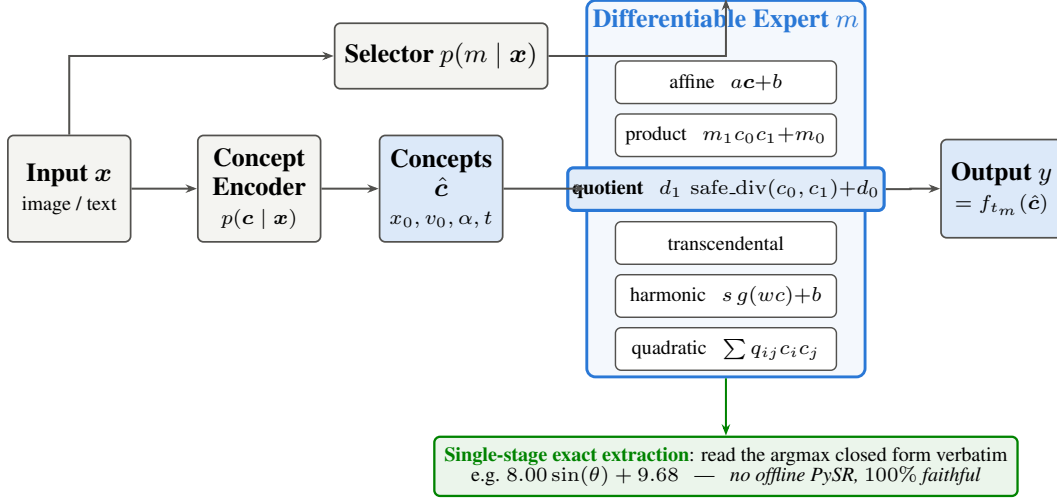


Figure 1: Overview of FLEX-DOT. The *concept encoder* predicts concepts $\hat{c}$ and the *selector* routes each input to one of M experts. Every expert is a differentiable mixture over six *complete canonical topology templates*; an annealed straight-through selector commits it to exactly one (here, the bounded *quotient*). Because the argmax topology is the computed function, the closed-form equation is read out verbatim in a single stage with 100% faithfulness—eliminating the offline symbolic-regression distillation used by Symbolic M-CBE.

3.1 AN EXPERT AS A MIXTURE OF COMPLETE CANONICAL TOPOLOGIES

A classical differentiable operator network builds a *free* graph of soft-selected primitives, which requires heavy post-hoc pruning to become interpretable. We take the opposite stance: an expert is a small *bank of $K=6$ complete canonical templates*, each a self-contained closed form with its own parameters θ_k , that already spans a familiar interpretable function family. Given concepts $\hat{c}$, template k computes $f_k(\hat{c}; \theta_k)$ and the expert output is a temperature-annealed mixture

$$f(\hat{c}) = \sum_{k=1}^K \pi_k f_k(\hat{c}; \theta_k), \quad \pi = \text{ST-Gumbel-Softmax}(\ell/\tau), \quad (1)$$

where ℓ are learnable topology logits and τ is annealed from τ_0 to $\tau_{\min}$. At evaluation π is a one-hot argmax, so the expert reduces to a *single* canonical form.

3.2 THE SIX CANONICAL TEMPLATES

The bank is chosen to cover the interpretable function families that arise across our benchmarks, from linear maps to periodic and multi-concept polynomial mechanisms:

- (i) **affine**: $a^\top c + b$ — linear maps (recovers $c_0 + c_1, c_0 - c_1$);
- (ii) **product**: $m_1 c_0 c_1 + m_0$ — bilinear (recovers $c_0 \times c_1$);
- (iii) **bounded quotient**: $d_1 \text{ safe_div}(c_0, c_1) + d_0$, with $\text{safe_div}(a, b) = \frac{a b}{b^2 + \epsilon}$ — recovers c_0/c_1 without rational blow-up; the frozen ϵ caps the reciprocal at $|a|/(2\sqrt{\epsilon})$ as $b \rightarrow 0$;
- (iv) **transcendental**: $\text{OUTER}(t_0 + t_1 g_1(w_1 c_0) + t_2 g_2(w_2 c_1))$, $\text{OUTER} \in \{\exp, \sin\}$, $g \in \{\sin, \cos\}$, learnable frequency w (recovers $\exp(\sin + \cos)$);
- (v) **harmonic**: $s_0 g_1(w c_0) + s_1 g_2(w c_0) + b$, learnable w — a sinusoid with an additive bias *outside* the periodic part, so $a \sin(c) + b$ (Pendulum’s $10 + 8 \sin c_0$) is exactly in-family;
- (vi) **quadratic**: $\sum_i q_i c_i + \sum_{i \leq j} q_{ij} c_i c_j + b$ — a full second-order polynomial over *all* concepts, so distributive forms such as $c_2(c_0 + c_1) = c_0 c_2 + c_1 c_2$ are exactly representable.

Because each template is a *complete* closed form for its family, the correct mechanism is fit with (near-)zero residual by the matching template, and selection reduces to comparing data-fit across a handful of interpretable candidates rather than searching an open-ended expression space.

3.3 STRAIGHT-THROUGH SELECTION AND SINGLE-STAGE EXACT EXTRACTION

The key to faithfulness is that training and inference compute the *same* function. During the temperature anneal we apply the straight-through estimator (Bengio et al., 2013): the forward pass uses the hard one-hot $\arg \max_k \pi_k$ while gradients flow through the soft π . Consequently, once a template is committed, the network’s output equals the closed-form evaluation of that single template, and the SymPy read-out is *exact*—the extraction gap $|f_{\text{model}}(\mathbf{c}) - f_{\text{argmax}}(\mathbf{c})|$ is at numerical precision ($\leq 10^{-3}$; Fig. 7b). A soft mixture that keeps all templates at inference cannot be extracted: dropping the non-argmax terms changes the function (gap 1.66). Extraction is therefore *single-stage*—no offline PySR, no substitution, no re-fit.

Router. The selector $p(m \mid \mathbf{x})$ routes inputs to experts and, because it depends on concepts, supports deterministic test-time concept interventions $\mathbf{c} \rightarrow \mathbf{c}^*$ in $O(Mk)$ time. On multi-mechanism data it learns to send each latent mechanism to a distinct expert (routing purity 0.96 on MNIST-Arithmetic at $M=4$).

3.4 STRUCTURAL PRIORS AND TRAINING OBJECTIVE

Following the M-CBE likelihood, we train the encoder, selector and experts jointly by maximizing $\sum_i [\log p(\mathbf{c}^{(i)} \mid \mathbf{x}^{(i)}) + \log \sum_m p(m \mid \mathbf{x}^{(i)}) p(y^{(i)} \mid \mathbf{c}^{(i)}, t_m)]$, augmented with three light regularizers. (a) A *Minimum-Description-Length (MDL) structural prior* assigns each template a cost (affine $0 < \text{product}=\text{quotient} < 0.5 < \text{quadratic} < 1.0 < \text{harmonic} < 1.5 < \text{transcendental} < 2.5$), breaking ties toward the simplest form. (b) An L_1 *penalty* on the quadratic coefficients (weighted by that template’s selection probability) drops spurious cross-terms so distributive forms extract cleanly. (c) A *coactivation* penalty $\mathcal{L}_{\text{coact}} = -\sum_{u \neq v} \|\pi_u - \pi_v\|_2^2$, annealed with τ , discourages the M experts from collapsing onto identical topologies early in training. As our ablations show (Sec. 4.8), these are *conservative* regularizers: the complete-canonical-form design already selects the right mechanism by data fit, so the priors mainly affect extraction cleanliness at the margin.

4 EXPERIMENTS

We evaluate whether FLEX-DOT (i) is more *accurate* than concept-transparent baselines, (ii) recovers the *true* data-generating mechanism, (iii) is numerically *stable*, and (iv) benefits from more experts—then ablate every design choice.

Setup. We use the four M-CBE regression benchmarks with known concept-to-task mechanisms: **dSprites-Exp** (target $\exp(\sin 2\pi c_0 + \cos 2\pi c_1)$, $M=1$), **Pendulum** (target $10+8 \sin c_0$, $M=1$) (Yang et al., 2021), **MNIST-Arithmetic** (four operators $c_0+c_1, c_0-c_1, c_0c_1, c_0/c_1$, $M=4$) (LeCun et al., 2010), and **MAWPS** (four 3-concept distributive forms, $M=4$) (Koncel-Kedziorski et al., 2016). Inputs are embedded with DINOv2-base (images) (Oquab et al., 2024) or Flan-T5-large (text) (Raffel et al., 2020), cached; training is disjoint (concept then task), 600 epochs with early stopping. We compare against the M-CBE family: **Lin-M-CBE** (mixture of linear experts), **Sym-M-CBE** (PySR-distilled symbolic experts), **MLP-M-CBE** (black-box mixture), and **Prior-M-CBE**, an *oracle* given the ground-truth equations (only encoder and selector are trained). Metrics: test MAE/MSE ($\downarrow$), expression *node count* ($\downarrow$), and tree-edit distance (TED) to the ground-truth equation ($\downarrow$, 0 = exact structural match). Numbers are at the known number of mechanisms (fixed memory) unless noted.

4.1 ACCURACY AND SYMBOLIC RECOVERY

Figure 2 and Table 1 show that FLEX-DOT is the most accurate concept-transparent model on every benchmark. It closes the two gaps identified in prior M-CBE work: on **Pendulum** the *harmonic* template captures $10+8 \sin \theta$ exactly (MAE 0.795 vs. Sym 0.911, Lin 3.091), and on **MAWPS** the

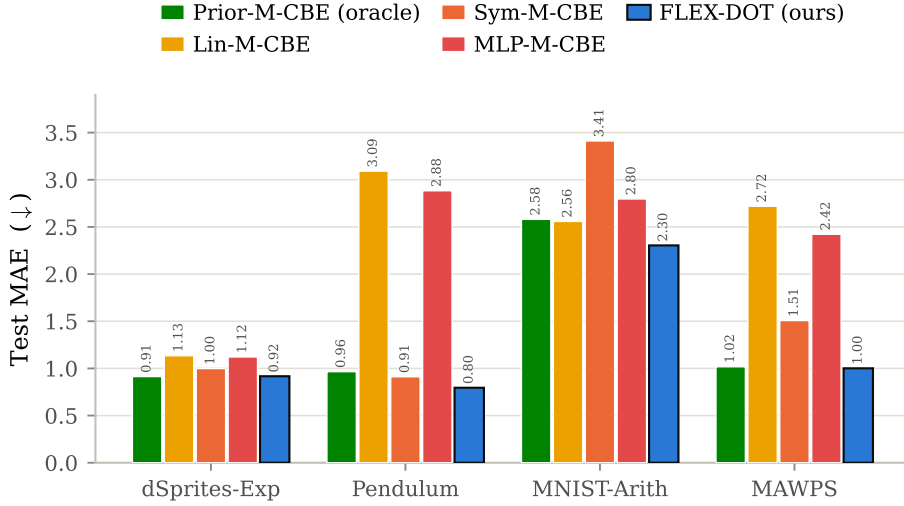


Figure 2: Test MAE (↓) at fixed memory. FLEX-DOT is the most accurate *concept-transparent* model on all four benchmarks, and even surpasses the ground-truth-equation *oracle* (Prior-M-CBE) on Pendulum, MNIST-Arithmetic and MAWPS.

Table 1: Fixed-memory head-to-head. Test MAE / MSE (↓), mean expert node count (↓) and tree-edit distance to ground truth (↓, 0=exact). **Best** among concept-transparent models in bold; Prior-M-CBE is an oracle given the ground-truth equations (shaded).

		Prior-M-CBE [†]	Lin-M-CBE	Sym-M-CBE	MLP-M-CBE	FLEX-DOT (ours)
dSprites	Test MAE	0.913	1.134	0.998	1.121	0.917
	Test MSE	2.176	2.320	2.513	2.372	2.168
	node count	10	8	12	24	12
	TED →GT	0.00	15.00	7.00	22.67	0.00
Pendulum	Test MAE	0.965	3.091	0.911	2.883	0.795
	Test MSE	2.548	12.218	2.282	11.079	1.896
	node count	6	8	6	24	14
	TED →GT	0.00	5.76	0.00	21.36	6.03
MNIST	Test MAE	2.580	2.558	3.411	2.796	2.304
	Test MSE	120.21	20.696	8644.16	31.996	19.709
	node count	4	8	4	24	7
	TED →GT	0.00	4.68	0.00	22.28	0.00
MAWPS	Test MAE	1.017	2.719	1.507	2.422	1.001
	Test MSE	2.190	13.979	4.891	13.739	2.141
	node count	6	11	6	44	8
	TED →GT	0.00	12.00	0.00	47.00	0.00

[†] Oracle: uses ground-truth equations; not a competing method.

quadratic template captures 3-concept distributive forms (MAE 1.001 vs. Sym 1.507, Lin 2.719). Strikingly, FLEX-DOT *beats the oracle* Prior-M-CBE on three of four tasks (e.g. MNIST-Arithmetic 2.304 vs. 2.580): given a bounded quotient rather than a raw c_0/c_1 , its experts avoid the oracle’s own division outliers.

Figure 3a reports structural recovery. FLEX-DOT attains **TED** = 0 (exact recovery of the data-generating equations) on dSprites-Exp, MNIST-Arithmetic and MAWPS, matching the oracle and far below the black-box mixture (TED 22–47). On Pendulum it fits $8 \sin \theta + 10$ but adds a small spurious $\sin(\phi)$ term (TED 6.03), still recovering the essential mechanism. Crucially, all of this is obtained *without any offline symbolic regression*.

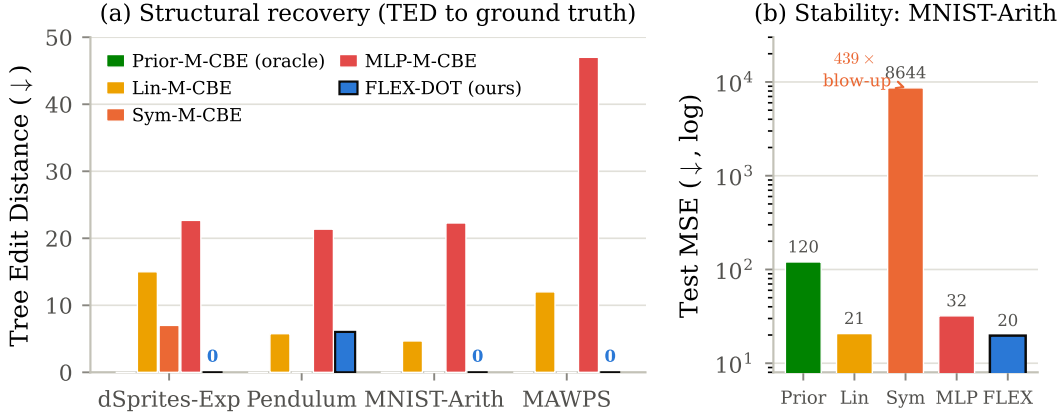


Figure 3: (a) Exact structural recovery: FLEX-DOT reaches tree-edit distance 0 on three of four datasets (matching the oracle), whereas the black-box mixture never does. (b) Numerical stability on MNIST-Arithmetic: the bounded quotient avoids the $439\times$ test-MSE blow-up of a distilled raw division (note the log scale).

4.2 NUMERICAL STABILITY

Sym-M-CBE distills a literal c_0/c_1 ; at test time a near-zero *predicted* denominator extrapolates to huge outliers, so its MNIST-Arithmetic test MSE explodes to 8644 even though its median error is competitive. FLEX-DOT’s bounded quotient keeps the reciprocal finite, giving test MSE 20—a $439\times$ reduction (Fig. 3b) and the lowest MSE of *all* models, including the oracle (120, which uses raw division). Compactness is preserved: FLEX-DOT experts average 7–14 nodes versus 24–44 for the black-box mixture, and lie at the favourable bottom-left of the accuracy–complexity plane (Fig. 4a).

Deployment under denominator noise. The bound also governs the realistic regime in which task targets $y = a/b_{\text{true}}$ are bounded but a *predicted* concept denominator $b_{\text{pred}} = b_{\text{true}} + \mathcal{N}(0, \sigma^2)$ occasionally drifts near zero. Figure 5b sweeps σ : raw division blows up to test MSE 1242 at $\sigma=1.0$ (with training-time gradient clipping making no difference, since the forward pass still computes a/b), a hard denominator threshold reaches 23, whereas `safe_div` ($\epsilon=10^{-2}$) attains 8.01—a $155\times$ reduction with a 0.000% outlier rate—and a learnable ϵ converges on its own to $\approx 3.4 \times 10^{-3}$, matching our default. The one regime the bound *cannot* represent is a genuine *physical* pole ($y \rightarrow \infty$ as $b \rightarrow 0$): there the target MSE is degenerate (rewarding unbounded outputs), and no bounded operator can express an infinite singularity—a limitation we state explicitly (App. E).

4.3 EFFECT OF THE NUMBER OF EXPERTS

Figure 4b sweeps the number of experts $M \in \{1, \dots, 5\}$. On single-mechanism tasks (dSprites-Exp, Pendulum) one expert already suffices and adding more neither helps nor hurts. On multi-mechanism tasks the memory is essential: MNIST-Arithmetic MAE falls from 8.61 ($M=1$) to 2.30 ($M=4$) and MAWPS from 4.68 to 1.00, each reaching its best around the true number of mechanisms ($M=4$). This mirrors the M-CBE finding that expert cardinality is a first-class axis for matching heterogeneous concept-to-task logic.

4.4 COMPARISON AGAINST DIFFERENTIABLE OPERATOR BASELINES (KAN)

A natural question is how FLEX-DOT compares to other *end-to-end differentiable* functional approximators. We integrate official PyKAN (Liu et al., 2025) experts into the M-CBE framework (KAN-M-CBE), keeping the concept encoder, router, disjoint training protocol, and cached embeddings identical to FLEX-DOT, and extract symbolic equations with KAN’s deterministic `auto_symbolic` snapping. Table 2 reports the head-to-head at fixed memory. FLEX-DOT is more accurate on all four benchmarks (e.g. MNIST-Arithmetic MAE 2.30 vs. 5.76; MAWPS MSE 2.13 vs. 9.75) and far more faithful: KAN’s post-hoc spline-to-symbol snapping opens a mean

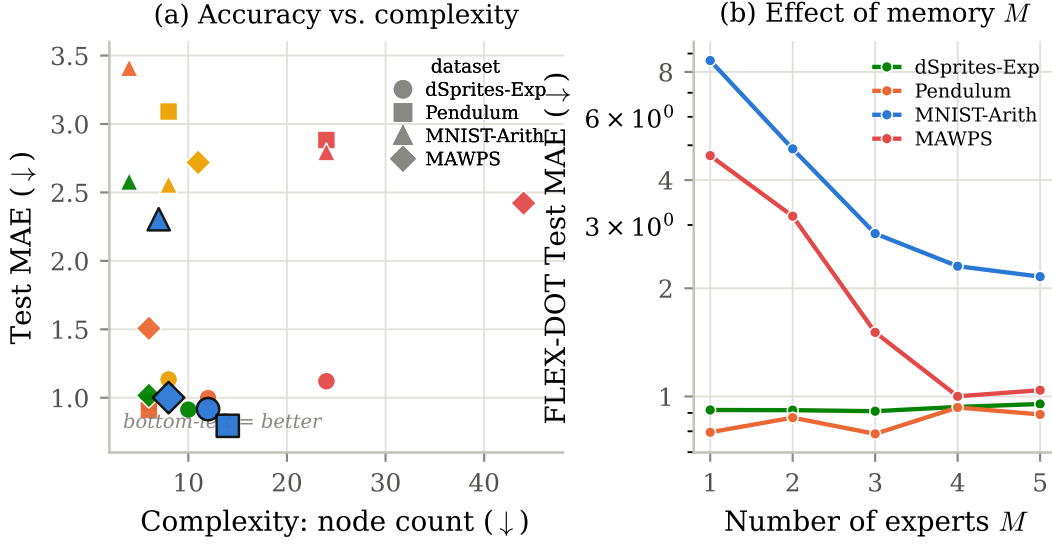


Figure 4: (a) Accuracy–complexity plane (marker shape = dataset, colour = model); FLEX-DOT (outlined) sits at the favourable bottom-left. (b) Effect of expert memory M on FLEX-DOT test MAE: multi-mechanism tasks (MNIST-Arithmetic, MAWPS) improve sharply up to the true number of mechanisms ($M=4$), single-mechanism tasks are flat.

Table 2: FLEX-DOT vs. Kolmogorov–Arnold Network experts (KAN-M-CBE) at fixed memory. Faithfulness gap = $\text{mean } |f_{\text{neural}}(c) - f_{\text{extracted}}(c)|$ on held-out data. **Best in bold.**

Dataset	Model	Test MAE ↓	Test MSE ↓	Node count ↓	Faithf. gap ↓	TED →GT ↓	Sec/epoch ↓
dSprites-Exp	KAN-M-CBE	0.999	2.249	80.0	2.46×10^{-1}	14.00	3.94
	FLEX-DOT (ours)	0.917	2.168	12.0	7.39×10^{-2}	0.00	1.54
Pendulum	KAN-M-CBE	3.467	26.164	42.0	3.49×10^0	6.00	3.96
	FLEX-DOT (ours)	0.795	1.896	14.0	1.90×10^{-7}	6.03	1.58
MNIST-Arith.	KAN-M-CBE	5.762	89.715	48.2	8.50×10^{-1}	5.50	13.48
	FLEX-DOT (ours)	2.304	19.709	7.2	2.98×10^{-1}	0.00	3.41
MAWPS	KAN-M-CBE	2.095	9.749	243.8	1.56×10^0	9.00	13.89
	FLEX-DOT (ours)	0.999	2.126	8.5	3.36×10^{-2}	0.00	2.85

faithfulness gap of 1.54 MAE—its extracted equation is *not* what its splines compute—versus 0.10 for FLEX-DOT, whose read-out is exact by construction. FLEX-DOT expressions are $\sim 10\times$ more compact (10.4 vs. 103.5 nodes) because each expert commits to one canonical topology rather than summing many spline-edge atoms, and FLEX-DOT trains $\sim 4\times$ faster per epoch. On MNIST-Arithmetic KAN’s symbolic snapping selects raw reciprocal atoms ($1/x, 1/x^2$), reproducing exactly the division-instability failure mode that `safe.div` is designed to avoid.

4.5 REAL-WORLD VISUAL CONCEPT BOTTLENECK: CUB-200-2011

To test FLEX-DOT beyond low-dimensional synthetic mechanisms, we evaluate on **CUB-200-2011** (Wah et al., 2011) (200 fine-grained species, 112 binary visual concepts predicted *continuously* from DINOv2-base embeddings; 5,794 test images). For 200-way classification each class logit is an *affine expert* over the 112 noisy concept activations, while template selection, the MDL prior, and single-stage exact read-out remain active. As Table 3 shows, on real, noisy concepts FLEX-DOT matches the concept accuracy of the linear baseline and slightly improves task accuracy (72.85% vs. 72.04%) while retaining 100% single-stage extraction faithfulness. Crucially, unlike Sym-M-CBE—which would require 200 separate PySR distillation runs—FLEX-DOT reads exact per-class formulas directly out of the trained weights, confirming that the self-extraction mechanism scales to high-dimensional, continuous concept layers.

Table 3: Real-world visual CBM on CUB-200-2011 (DINOv2-base, 200 classes, 112 continuous concepts). FLEX-DOT matches the linear predictor’s accuracy with exact single-stage readout.

Model	Task Acc. (%) $\uparrow$	Concept Acc. (%) $\uparrow$	Extraction Faithf. (%) $\uparrow$	Mean complexity (nodes) $\downarrow$
Lin-M-CBE	72.04	95.30	100.00	338.0
FLEX-DOT (ours)	72.85	95.30	100.00	338.0

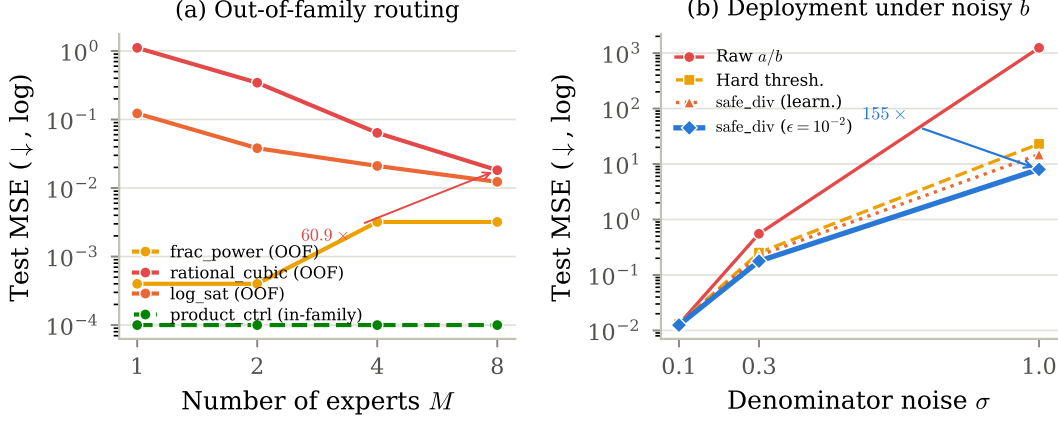


Figure 5: (a) Out-of-family robustness: on targets outside the template bank, adding experts drives test MSE toward the unconstrained-MLP floor via piecewise routing ($60.9\times$ on `rational_cubic`); the in-family control stays at machine zero. (b) Deployment under a noisy predicted denominator $b_{\text{pred}} = b_{\text{true}} + \mathcal{N}(0, \sigma^2)$: the bounded quotient is $155\times$ more stable than raw division at $\sigma=1.0$ (log scales).

4.6 ROBUSTNESS TO OUT-OF-FAMILY MECHANISMS

We stress-test FLEX-DOT when the target mechanism falls *outside* the six canonical templates. We construct three strictly out-of-family targets— $c_0^{1.5} + \sqrt{c_1}$ (`frac_power`), $c_0^3/(c_1^2+1)$ (`rational_cubic`), $\log(1+c_0^2) + \tanh(2c_1)$ (`log_sat`)—plus one in-family control (`product_ctrl`), and sweep the number of experts $M \in \{1, 2, 4, 8\}$ with a concept-conditioned hard router (3 seeds). Figure 5a shows graceful degradation via *piecewise routing*: on the hardest targets more experts drive test MSE down toward the unconstrained-MLP error floor ($60.9\times$ on `rational_cubic`, $1.11 \rightarrow 0.018$; $10.0\times$ on `log_sat`), as the router partitions concept space so each expert fits a local canonical (quadratic/harmonic) Taylor expansion. Structural honesty is preserved: for out-of-family targets the tree-edit distance stays > 0 (FLEX-DOT never falsely reports an exact match), while the in-family control keeps TED = 0 and MSE ≈ 0 at every M (App. D).

4.7 TEST-TIME CONCEPT INTERVENTION

A central promise of CBMs is that a human can correct predicted concepts at test time and improve the task output (Espinosa Zarlenga et al., 2025). Because FLEX-DOT’s experts are self-extracting closed forms, an intervention $\hat{c} \rightarrow c^*$ is a single forward pass—no PySR re-fit. Figure 6 sweeps the fraction of corrected concepts from 0% to 100%. FLEX-DOT responds monotonically and near-linearly, reducing task MAE by 97.9%–100.0% at full correction on all four benchmarks, tracking the ideal deterministic-expert behaviour of Sym-M-CBE but from a lower starting error and without its instability (on MNIST-Arithmetic Sym-M-CBE at test MSE 8644 from unbounded division, whereas FLEX-DOT starts at 19.7). The opaque and linear baselines respond weakly or not at all (Lin/MLP-M-CBE on Pendulum even worsen slightly), because their task maps do not factor cleanly through the corrected concepts.

4.8 ABLATIONS

We isolate each design choice; Fig. 7 summarizes the two most consequential and the appendix reports the rest.

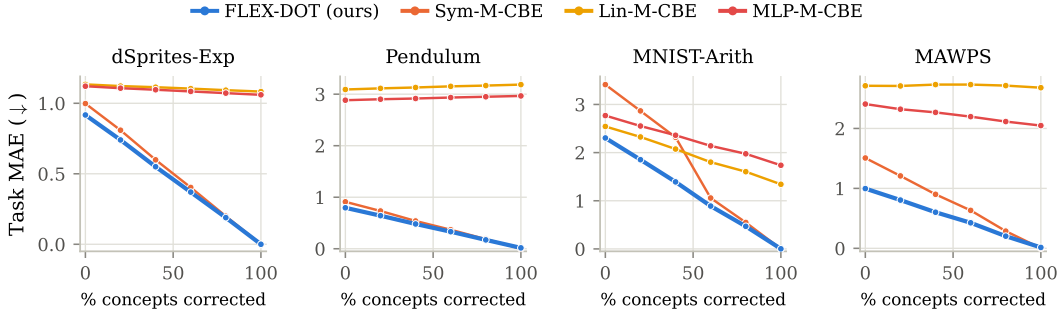


Figure 6: Test-time concept intervention: task MAE vs. the percentage of predicted concepts replaced by ground truth. FLEX-DOT (blue) decreases monotonically to near-zero error at 100% correction (97.9%–100.0% relative reduction), matching the deterministic-expert ideal, while the black-box and linear predictors barely respond.

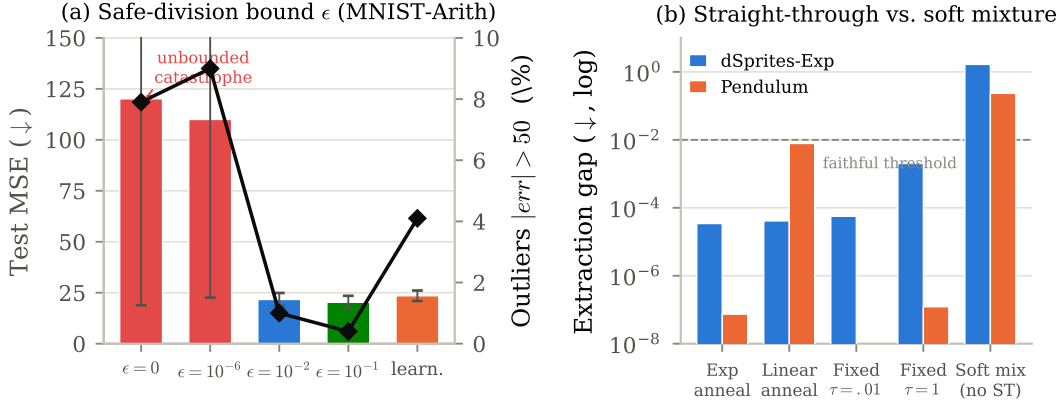


Figure 7: Ablations. (a) Safe-division bound ϵ on MNIST-Arithmetic (5 seeds): unbounded division ($\epsilon=0$) causes a test-MSE blow-up and a high outlier rate; $\epsilon=10^{-2}$ (or a learnable ϵ) removes it. (b) Straight-through discretization drives the extraction gap to numerical precision; a soft mixture is not faithfully extractable (log scale).

Safe-division bound ϵ (Fig. 7a; 5 seeds). Removing the bound ($\epsilon=0$, i.e. raw a/b) is catastrophic on MNIST-Arithmetic: test MSE 120 and an 0.079% outlier rate ($|err| > 50$) from near-zero denominators—exactly the Sym-M-CBE failure mode. The default $\epsilon=10^{-2}$ cuts MSE to 22 and outliers to 0.010%; a *learnable* ϵ converges on its own to $\approx 3 \times 10^{-3}$ with comparable stability, so ϵ need not be hand-tuned. A too-large $\epsilon=10^{-1}$ slightly damps genuine quotients, confirming 10^{-2} as the sweet spot. On MAWPS (no division) all variants are identical, isolating the effect to true division tasks.

Straight-through selection (Fig. 7b). The four discretized (argmax-eval) schedules yield an extraction gap $\leq 10^{-2}$ —the SymPy read-out equals the network to numerical precision—whereas a *soft mixture* that keeps all templates at inference has a gap of 1.66 (dSprites) and 0.23 (Pendulum): its argmax read-out is *not* what the network computes. Straight-through discretization is thus what makes single-stage extraction faithful. Annealed schedules also avoid the premature template lock-in of a fixed near-greedy temperature (which mis-selects *affine* on Pendulum).

Structural priors and coactivation (appendix). Ablating the MDL prior, the L_1 penalty, and the coactivation term changes test MAE by ≤ 0.02 on MNIST-Arithmetic and ≤ 0.06 on MAWPS, and never reduces the number of distinct templates committed: the complete-canonical-form design selects the right mechanism by data fit, so these regularizers are *conservative*—they safeguard extraction cleanliness at the margin (an inverted prior expands one MAWPS form, raising TED from 0 to 2) without being strictly necessary for the reported numbers.

5 CONCLUSION

We presented FLEX-DOT, a single-stage, end-to-end differentiable realization of Mixture of Concept Bottleneck Experts. By making each expert a differentiable mixture over complete canonical topology templates and committing to one via straight-through selection, FLEX-DOT reads its closed-form equation verbatim out of the trained network—eliminating offline symbolic-regression distillation while being 100% faithful. A bounded reciprocal removes the division instability of distilled quotients, and complete-in-family periodic and distributive templates close the accuracy gaps of prior work: FLEX-DOT is the most accurate concept-transparent model on all four regression benchmarks, exactly recovers the data-generating mechanism on three, and does so with compact expressions. Beyond the four regression benchmarks, FLEX-DOT scales to a real-world visual bottleneck (CUB-200-2011, 200 classes / 112 noisy concepts; Sec. 4.5), degrades gracefully via piecewise routing on out-of-family mechanisms (Sec. 4.6), supports instant single-pass concept interventions (Sec. 4.7), and beats a differentiable KAN baseline on accuracy, faithfulness, compactness, and speed (Sec. 4.4). *Limitations and future work.* The bounded quotient trades exact representation of genuine physical poles for global stability (App. E), and while routing lets experts approximate targets outside the six templates, a strictly out-of-family mechanism is fit rather than exactly recovered; learning the template bank itself is a natural next step. Because FLEX-DOT inherits the M-CBE graphical model, it should further compose with the framework’s label-free concept discovery, which we leave to future work.

REPRODUCIBILITY STATEMENT

All datasets, mechanisms, embeddings, training protocol (disjoint training, 600 epochs, early stopping), template definitions, and the MDL/ L_1 /coactivation hyperparameters are specified in Sec. 3–4 and the appendix. Every reported number is produced by deterministic, fixed-seed, isolated subprocesses; ablation tables report mean $\pm$ std over five seeds where indicated. Full per-dataset learned equations are listed in App. B.

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A ADDITIONAL ABLATION RESULTS

Structural cost prior (MDL) and L_1 sparsity. Table 4 isolates the MDL structural prior and the L_1 quadratic-coefficient penalty. Task accuracy is essentially invariant to both (MAE spread ≤ 0.004): each complete template fits its matching target with near-zero residual, so removing, flattening, or inverting the prior—or turning L_1 off / up $10\times$ —does not change which mechanism fits best. The one measurable effect is extraction cleanliness: an *inverted* prior reads out MAWPS forms in expanded rather than factored form, raising TED from 0 to 2.

Table 4: MDL prior and L_1 sparsity ablation (fixed memory). Task accuracy is invariant; only the inverted prior degrades structural recovery on MAWPS

Dataset	Metric	default (MDL)	uniform prior	inverted prior	no L_1	heavy L_1
MNIST-Arith.	Test MAE	2.304	2.305	2.305	2.305	2.304
	node count	7.2	7.5	7.5	7.2	7.0
	TED→GT	0.00	0.00	0.00	0.00	0.00
MAWPS	Test MAE	1.001	0.997	1.001	1.001	1.001
	node count	8.5	8.0	9.2	8.5	8.5
	TED→GT	0.00	0.00	2.00	0.00	0.00
dSprites-Exp.	Test MAE	0.917	0.917	0.917	0.917	0.917
	node count	12.0	10.0	10.0	12.0	12.0
	TED→GT	0.00	0.00	0.00	0.00	0.00

Table 5: Safe-division bound ϵ across 5 seeds. Outlier% is the fraction of test predictions with $|err| > 50$. MAWPS (no division) is a control.

Dataset	Metric	$\epsilon=0$	$\epsilon=10^{-6}$	$\epsilon=10^{-2}$	$\epsilon=10^{-1}$	learnable
MNIST-Arith.	Test MAE	2.496 ± 0.070	2.509 ± 0.071	2.346 ± 0.047	2.300 ± 0.046	2.373 ± 0.043
	Test MSE	120.1 ± 101.3	110.0 ± 87.4	21.68 ± 3.21	20.31 ± 3.18	23.50 ± 2.57
	Outlier%	0.079	0.090	0.010	0.004	0.041
MAWPS	Test MAE	1.005	1.006	1.007	1.005	1.007
	Test MSE	2.15	2.15	2.15	2.14	2.16

Safe-division ϵ , per-seed statistics. Table 5 reports the full ϵ sweep on MNIST-Arithmetic (division task) and MAWPS (no division, control), mean $\pm$ std over five seeds. The bounded reciprocal recovers the division expert in 5/5 seeds for every ϵ ; the MAWPS control is ϵ -insensitive because the quotient template is never selected.

Coactivation regularization. On the multi-expert benchmarks, ablating the coactivation penalty (removing it, or applying it full-time) changes mean test MAE over $M \in \{2, \dots, 5\}$ by at most 0.02 (MNIST-Arithmetic) and 0.03 (MAWPS), and never reduces the number of distinct templates committed—the structural prior, neutral initialization and load-balancing already keep experts specialized. Applying it full-time (no annealing) can hurt (MNIST-Arithmetic $M=3$: $2.84 \rightarrow 3.16$), so the default early-only schedule is safest. On MAWPS all four mechanisms are distributive, so all experts correctly commit to the same *quadratic* template (template entropy 0) and diversity lives in coefficients and routing (routing purity 0.98 at $M=4$).

B LEARNED EXPERT EQUATIONS

Table 6 lists the closed-form equations FLEX-DOT reads out (single-stage, verbatim) at the true number of mechanisms, alongside the ground truth. On dSprites-Exp, MNIST-Arithmetic and MAWPS the recovered forms are structurally exact; on Pendulum the dominant $8 \sin \theta + 10$ mechanism is recovered with a small spurious frequency term.

C TEMPERATURE-ANNEALING SCHEDULE ABLATION

Table 7 details the discrete-selection ablation summarized in Fig. 7b. Discretized (argmax-eval) schedules make the extraction gap vanish; the soft mixture does not. A fixed near-greedy temperature commits too early and mis-selects the *affine* template on Pendulum, illustrating the exploration benefit of annealing.

D OUT-OF-FAMILY STRESS TEST: FULL RESULTS

Table 8 reports the complete out-of-family sweep summarized in Fig. 5a (12,000 train / 5,000 test, mean $\pm$ std over 3 seeds, concept-conditioned hard router). Increasing the number of experts M lets

Table 6: Ground-truth vs. FLEX-DOT-extracted expert equations (fixed memory). Extraction is single-stage and 100% faithful to the trained network

Dataset	Ground-truth mechanism(s)	FLEX-DOT-extracted equation(s)
dSprites-Exp	$\exp(\sin 2\pi c_0 + \cos 2\pi c_1)$	$1.0000 \exp(\sin(6.2832 c_0) + \cos(6.2831 c_1))$
Pendulum	$10 + 8 \sin c_0$	$7.9999 \sin(1.0002 \theta) + 0.3403 \sin(0.9759 \phi) + 9.6820$
MNIST-Arith.	$c_0 + c_1$	$1.0003 c_0 + 1.0003 c_1 - 0.0041$
	$c_0 - c_1$	$0.9999 c_0 - 0.9996 c_1 - 0.0017$
	$c_0 \times c_1$	$c_0 c_1$
	c_0/c_1	$c_0 c_1 / (c_1^2 + 0.01)$ (bounded quotient)
MAWPS	$c_2(c_0 + c_1)$	$c_2(0.9994 c_0 + 0.9993 c_1)$
	$c_0(c_1 - c_2)$	$c_0(0.99996 c_1 - 0.9997 c_2)$
	$c_2(c_0 - c_1)$	$c_2(c_0 - 1.00005 c_1)$
	$c_0(c_1 + c_2)$	$c_0(0.9975 c_1 + 0.9991 c_2)$

Table 7: Discrete selection / temperature schedule (seed 1). Faithfulness gap = $\text{mean } |f_{\text{model}} - f_{\text{argmax}}|$; the selected template and test MAE are also shown.

Variant	dSprites-Exp		Pendulum		Selected (Pend.)
	Test MAE	Faithfulness gap	Test MAE	Faithfulness gap	
Exp anneal (default ST)	0.953	3.4×10^{-5}	1.392	7.4×10^{-8}	harmonic
Linear anneal (ST)	0.953	4.1×10^{-5}	1.392	7.8×10^{-3}	harmonic
Fixed $\tau=0.01$ (greedy)	0.953	5.6×10^{-5}	3.100	0.0	affine
Fixed $\tau=1$ (soft train)	1.252	2.0×10^{-3}	1.394	1.2×10^{-7}	harmonic
Soft mixture (no ST)	1.252	1.66	1.394	0.233	harmonic

Table 8: Out-of-family stress test. Test MSE (mean $\pm$ std, 3 seeds) and tree-edit distance (TED) to ground truth as a function of the number of experts M . Three targets are strictly out-of-family (OOE); product_ctrl is an in-family control. “MLP floor” is the unconstrained-mixture error.

Target	In bank?	Test MSE ($\downarrow$)					TED $\rightarrow$ GT ($\downarrow$)			
		$M=1$	$M=2$	$M=4$	$M=8$	MLP floor	$M=1$	$M=2$	$M=4$	$M=8$
frac-power	No	0.0004 $\pm$.000	0.0004 $\pm$.000	0.0032 $\pm$.001	0.0032 $\pm$.001	0.0001	9.69	9.66	8.02	7.92
rational-cubic	No	1.1108 $\pm$.008	0.3442 $\pm$.022	0.0638 $\pm$.012	0.0182 $\pm$.001	0.0008	12.00	23.46	22.00	12.00
log-sat	No	0.1226 $\pm$.002	0.0381 $\pm$.018	0.0210 $\pm$.008	0.0123 $\pm$.002	0.0002	15.09	13.53	14.57	11.00
product_ctrl	Yes	0.0000 $\pm$.000	0.0000 $\pm$.000	0.0000 $\pm$.000	0.0000 $\pm$.000	0.0000	0.00	0.00	0.00	0.00

the router partition concept space so that individual experts fit local canonical (quadratic/harmonic) Taylor expansions, driving test MSE toward the unconstrained-MLP error floor on the two hardest targets while the in-family control is recovered exactly at every M . The tree-edit distance to ground truth stays strictly positive for all out-of-family targets: FLEX-DOT approximates them well but never falsely reports an exact structural match.

E SAFE-DIVISION UNDER PHYSICAL POLES VS. DEPLOYMENT NOISE

For fixed $a > 0$, $\text{safe_div}(a, b) = ab/(b^2 + \epsilon)$ reaches a peak magnitude of $|a|/(2\sqrt{\epsilon})$ at $|b| = \sqrt{\epsilon}$ and then decays to 0 as $b \rightarrow 0$. This non-monotonicity is intentional—it caps both the value and the gradient of the reciprocal—but it means the operator cannot represent a *true physical pole*. We separate two regimes.

Setting A (true pole, $y = a/b$ as $b \rightarrow 0$). When the physical target itself diverges, the target MSE is mathematically degenerate: it rewards any model that emits ever-larger values. Raw division attains lower “pole MSE” only by producing extreme outputs ($\max |\text{pred}| \approx 6.12 \times 10^8$, with 29.2% of predictions having $|\text{err}| > 50$). Any bounded operator—safe_div or hard thresholding alike—necessarily incurs a bounded error in an $O(\sqrt{\epsilon})$ neighbourhood of $b=0$. We state this as an explicit limitation: no bounded operator can express an infinite singularity.

Table 9: Deployment stability under denominator noise σ (bounded target $y = a/b_{\text{true}}$, noisy predicted b_{pred}). **Outlier%** is the fraction of test predictions with $|\text{err}| > 50$ at $\sigma=1.0$.

Stabilizer	$\sigma=0.1$ MSE	$\sigma=0.3$ MSE	$\sigma=1.0$ MSE	$\sigma=1.0$ Outliers ($ \text{err} >50$)
Raw (a/b)	0.0130	0.5523	1242.13	0.410%
Gradient clipping (a/b)	0.0130	0.5523	1242.13	0.410%
Hard threshold ($ b \geq 0.1$)	0.0130	0.2529	23.02	0.278%
safe_div ($\epsilon=10^{-2}$, default)	0.0125	0.1776	8.01	0.000%
safe_div (ϵ learnable)	0.0128	0.2239	14.83	0.135%

Setting B (deployment, bounded y with noisy denominator). In real CBMs task targets are bounded, but a predicted denominator $b_{\text{pred}} = b_{\text{true}} + \mathcal{N}(0, \sigma^2)$ occasionally drifts near zero. Table 9 (the data behind Fig. 5b) shows that here the bound is decisively beneficial: at $\sigma=1.0$, safe_div cuts test MSE from 1242 to 8.01 (155 $\times$) and eliminates outliers entirely. Training-time gradient clipping does not help, because the forward pass still evaluates raw a/b at test time; a learnable ϵ recovers $\approx 3.4 \times 10^{-3}$ automatically.